

THE FREE TWIN-PRIME BOST–CONNES SYSTEM IS UNCONDITIONALLY SUPERCRITICAL: RESOLVING THE PHASE DICHOTOMY VIA EXACT PARTIAL SUMS

RUQING CHEN

ABSTRACT. For the free (Toeplitz–Cuntz) variant of the localized Bost–Connes system of a prime set S , the partition series is geometric in $L_S(\beta) = \sum_{p \in S} p^{-\beta}$, so the phase dichotomy of Paper V is decided by the fugacity at the classical critical point: Regime (S), $\beta_c^{\text{free}} > 1$, holds if and only if $L_S(1) > 1$. For the twin primes this comparison had resisted decision because the natural route — the Brun identity $L_S(1) = \frac{1}{2}B_2 + \sum_{p \in S} \frac{1}{p(p+2)}$ — feeds on the value of Brun’s constant, and the familiar $B_2 \approx 1.902160$ is a Hardy–Littlewood *extrapolation*, not a theorem. We resolve the dichotomy unconditionally by the blunt instrument the problem actually admits: partial sums of a positive series are rigorous lower bounds, and a segmented sieve to 1.1×10^{10} (29,889,853 twin pairs) gives

$$\sum_{\substack{p \leq 1.1 \times 10^{10} \\ p, p+2 \in \mathcal{P}}} \frac{1}{p} = 1.00195935 > 1.$$

Hence $L_{S_{\text{twin}}}(1) > 1$ and the free twin-prime system is in Regime (S): $\beta_c^{\text{free}} > 1$, unconditionally, so on the interval $(1, \beta_c^{\text{free}})$ the classical system is already in its broken phase while the free system admits no equilibrium at all. The Hardy–Littlewood rate predicts 1.0017 at this height, a cross-check, and the conditional value $\beta_c^{\text{free}} \approx 1.009$ is recorded as an estimate only. Two methodological remarks close the paper: occupancy of residue classes does not imply divergence of the per-character detecting series (the gap that keeps the Chen uniqueness problem open is Chen’s theorem in arithmetic progressions), and average correlation supplied by Rankin–Selberg poles does not touch the pointwise parity barrier: metaphor is not proof.

1. THE DICHOTOMY AND THE IDENTITY

Let $S \subseteq \mathcal{P}$ and let $\mathcal{TC}(S)$ be the free variant of Paper V: the generating isometries have *orthogonal* ranges, one for each $p \in S$, so the gauge-invariant counting runs over the free monoid on S and the Gibbs series is geometric,

$$Z_S^{\text{free}}(\beta) = \sum_{n \geq 0} L_S(\beta)^n, \quad L_S(\beta) = \sum_{p \in S} p^{-\beta}.$$

Thus a free equilibrium exists iff $L_S(\beta) < 1$, and Paper V’s trichotomy at the classical critical point $\beta_c(S_{\text{twin}}) = 1$ (Brun-summable, closed classical phase, Paper XXIV) reads: *Regime (S)* if $L_S(1) > 1$, with β_c^{free} the unique root of $L_S(\beta) = 1$ in $(1, \infty)$; *Regime (B)* if $L_S(1) \leq 1$. The decision is a single comparison with 1.

For $S = S_{\text{twin}}$ (smaller members p of twin pairs), the elementary identity $\frac{1}{p+2} = \frac{1}{p} - \frac{2}{p(p+2)}$ summed over pairs gives, with Brun’s constant $B_2 = \sum_{p, p+2 \in \mathcal{P}} (\frac{1}{p} + \frac{1}{p+2})$ (finite by Brun’s sieve

Date: 6 September 2026.

Paper XXXV of the series on localized Bost–Connes systems. The framework is that of [1], cited by DOI; Papers V and XXIV of the series are referred to by numeral, and the facts used are restated in place. Script and output in `code/twin-regime/` of the repository.

[2]),

$$(1) \quad L_{S_{\text{twin}}}(1) = \frac{1}{2}B_2 + \sum_{p \in S_{\text{twin}}} \frac{1}{p(p+2)}, \quad \sum_{p \in S_{\text{twin}}} \frac{1}{p(p+2)} = 0.10798 \dots$$

(the second series converges rapidly and is evaluated by direct summation). The identity cannot decide the dichotomy by itself: the familiar value $B_2 \approx 1.9021605$ [3] is the computed partial sum *plus a Hardy–Littlewood extrapolation of the tail*; unconditionally one only knows the partial sums, ≈ 1.83 at the computed heights. Conditionally, (1) gives $L_S(1) \approx 0.9511 + 0.1080 = 1.059$; but a conditional inequality decides nothing.

2. UNCONDITIONAL RESOLUTION VIA SEGMENTED SIEVING

The problem admits a blunt unconditional instrument: $L_S(1)$ is a series of positive terms, so every partial sum is a rigorous lower bound, and the comparison with 1 only requires some partial sum to cross it.

Theorem 2.1. $\sum_{p \leq 1.1 \times 10^{10}, p, p+2 \in \mathcal{P}} \frac{1}{p} = 1.00195935 \dots > 1$, the sum running over 29,889,853 twin pairs enumerated by a deterministic segmented sieve of Eratosthenes (blocks of 2×10^8 , base primes to $\sqrt{1.1 \times 10^{10}}$; script and output archived in `code/twin-regime/`). Consequently $L_{S_{\text{twin}}}(1) > 1$ unconditionally: the free twin-prime system is in Regime (S), $\beta_c^{\text{free}}(S_{\text{twin}}) > 1$, and on $(1, \beta_c^{\text{free}})$ the classical localized system is in its closed broken phase while the free system admits no KMS state at all.

Proof. The enumeration is exact and finite; positivity does the rest. Floating-point error is dominated by 3×10^7 additions of magnitude $\leq \frac{1}{3}$ in double precision, orders below the margin 1.96×10^{-3} . As a cross-check, the Hardy–Littlewood rate $2C_2/\log^2 t$ integrated from the previous checkpoint ($\sum_{p \leq 3 \times 10^8} = 0.991422$) predicts 1.0017 at 1.1×10^{10} , consistent. $\square$

Remark 2.2. The precise root of $L_S(\beta) = 1$ depends on the full tail and is therefore tail-conditional; under Hardy–Littlewood it is $\beta_c^{\text{free}} \approx 1.009$ (and ≈ 1.211 for the Chen family). These are estimates, not assertions: the theorem above is exactly the part of the phase diagram that a finite computation can own.

3. METHODOLOGICAL REMARKS: WHY HEURISTICS DO NOT BYPASS THE SIEVE

Remark 3.1 (Occupancy does not imply divergence). High-temperature KMS uniqueness for a family S requires, by [Main, Thm 3.9], the divergence of the *per-character* detecting series $\sum_{p \in S, \chi(p) \neq 1} p^{-\beta}$ for every nontrivial Dirichlet χ . For the Chen primes, the full series diverges unconditionally for $\beta < 1$ (Chen’s lower bound plus Abel summation), and the occupied classes generate every $(\mathbb{Z}/M)^\times$ (Paper XXIV); but occupancy is a statement about *which* classes carry primes, not about *how much* mass each class carries, and upper bounds in progressions cannot split a divergent series into per-class divergences ($\infty - \infty$). What is needed, and remains the named gap, is Chen’s theorem *in arithmetic progressions*: a log-power lower bound in every class outside each kernel. Three proposals in this project’s history have foundered on exactly this step; it is a sieve problem, and it does not dissolve under relabeling.

Remark 3.2 (Metaphor is not proof). Rankin–Selberg poles at $\Re s = 1$ encode *average* pair correlation of automorphic coefficients; the parity barrier of sieve theory is a statement about the *pointwise* indistinguishability of integers with an even versus odd number of prime factors under sieve weights. Passing the twin condition through symmetric-power L -functions, gauge-flow readings of the circle method, or fermionic vacuum imagery changes the vocabulary and not the

obstruction: no functional-analytic reformulation in this series has produced, nor could produce, the missing pointwise lower bound. The honest unconditional content available today for the twin primes is Theorem 2.1; everything sharper is conditional, and is labeled so.

REFERENCES

- [1] R. Chen, *KMS state uniqueness and phase transitions in localized Bost–Connes systems*, preprint, 2026, <https://doi.org/10.5281/zenodo.22152101>.
- [2] H. Halberstam, H.-E. Richert, *Sieve Methods*, Academic Press, 1974.
- [3] T. R. Nicely, *Enumeration to 10^{14} of the twin primes and Brun’s constant*, Virginia J. Sci. 46 (1995), 195–204.

GUT GEOSERVICE INC., MONTREAL, CANADA
Email address: ruqing@hotmail.com